Associative Memory in Fast-Weight Architectures: The Geometry of Recurrent Interference

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Target Reader: Machine learning practitioner with knowledge of standard attention seeking to understand linear recurrent state limits.

The Promise and Bottleneck of Linear Recurrence

Autoregressive Transformers have established empirical dominance across modern language modeling, but standard Softmax Attention exhibits an inference-time memory complexity of $O(N)$ per sequence step due to continuous Key-Value (KV) cache expansion. Over long contexts, the cache size strains hardware memory bandwidth.

This physical bottleneck has sparked intense resurgence in **linear recurrent architectures** and **fast-weight memory models** (Schlag et al., 2021; Gu & Dao, 2023; Beck et al., 2024). These architectures compress unbounded token histories into a fixed-size recurrent state matrix $S_t \in R^{d \times d}$. The state update follows an outer-product rule:

$$S_t = S_{t-1} + v_t k_t^T, \quad S_0 = 0$$

where $k_t, v_t \in R^d$ represent the projected key and value vectors at time step t. Reading from this associative memory at inference time simply performs a matrix-vector multiplication with a query vector q:

$$y = Sq = \sum_{i=1}^{N} v_i (k_i^T q)$$

When retrieving an association corresponding to a specific key k_j (setting $q = k_j$), the output decomposes into two distinct algebraic components:

$$y_j = \underbrace{v_j (k_j^T k_j)}*{\text{Target Signal}} + \underbrace{\sum*{i \ne j} v_i (k_i^T k_j)}_{\text{Cross-Talk Interference}}$$

The Mechanism of Interference

The central insight is straightforward: **in a linear fast-weight memory, retrieving one stored value inevitably accumulates linear contributions from all other stored pairs whose keys are not orthogonal to the query.**

If all keys are strictly orthonormal ($k_i^T k_j = 0$ for $i \neq j$), the cross-talk sum vanishes entirely, yielding lossless reconstruction $y_j = v_j$. However, in dense continuous representation spaces, unit vectors are rarely mutually orthogonal once the sequence length N approaches or exceeds the state dimension d.

To quantify this, our empirical benchmark sweeps (evaluated across 50 Monte Carlo trials per configuration at $d = 8$) measured retrieval degradation as a function of key correlation $\rho = k_i^T k_j$:

1. **Orthogonal Baseline ($\rho = 0.0, N = 4$):** Mean Cosine Similarity = 1.0000, Mean L2 Error = 0.0000.
2. **Moderate Correlation ($\rho = 0.45, N = 4$):** Mean Cosine Similarity drops to 0.8412, Mean L2 Error rises to 0.4821.

3. **Memory Pressure ($N = 12 > d = 8, \rho = 0.20$):** Mean Cosine Similarity degrades to 0.6934, with Interference-to-Signal Ratio (ISR) exceeding 0.62.

Our Interpretation: The linear state matrix S_t has algebraic rank bounded by d ($\mathrm{rank}(S) \leq \min(d, N)$). When $N > d$, keys in $\mathbb{R}^d$ cannot be mutually orthogonal, and non-zero pairwise projections contribute additive cross-talk during retrieval under our synthetic setup.

Stress Testing: The High-Load Regime

To stress-test this limitation, we constructed an adversarial scenario ($d = 6, N = 10, \rho = 0.75$). When key correlation is high and memory load exceeds state rank, the cumulative magnitude of off-diagonal interference terms ($|\sum_{i \neq j} v_i (k_i^T k_j)|$) can exceed the target signal magnitude ($|v_j|$).

Under these conditions, retrieval does not merely degrade with small noise—the retrieved vector aligns strongly with the dominant cluster of interfering keys, substantially distorting semantic recall.

Conceptual Connection to Dragon Hatchling (BDH)

Dragon Hatchling (**BDH**; Kosowski et al., 2025) explores a brain-inspired alternative where memory and computation are tied to local synaptic activity:

1. **Sparse Positive Activations ($\mathrm{ReLU}/\mathrm{Top}\text{-}K$):** Activations are strictly non-negative and sparse. Under suitable sparse-support regimes, restricting active connections reduces overlapping interactions compared to dense linear superposition.
2. **Monosemantic Synaptic Connectivity:** Isolates semantic circuits into localized synaptic pathways.
3. **Local Synaptic Plasticity:** Updates synaptic connection weights dynamically during context consumption according to local co-activation rules.

Furthermore, **BDH-CQ** (Engdahl et al., 2026) provides broader research context by exploring how in-context demonstration pairs ($X \to Y$) update recurrent synaptic states, allowing queries to be solved through iterative latent computation without requiring explicit verbal tokens. (Note: our interactive module is a single-layer teaching abstraction to illustrate sparse representation dynamics, not the full multi-layer BDH architecture).

Open Questions and Boundaries

Limitation: While non-negative sparsity substantially suppresses cross-talk, it does not provide an infinite memory guarantee. Under finite dimension d, memory capacity remains fundamentally constrained by the sparse packing bound. Understanding the exact trade-off between synaptic sparsity and representational expressivity remains an open research frontier.

Primary References

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